OPENING | BUILD INDONESIA | AEROSPACE & DEFENSE

ATC FLOWLENS

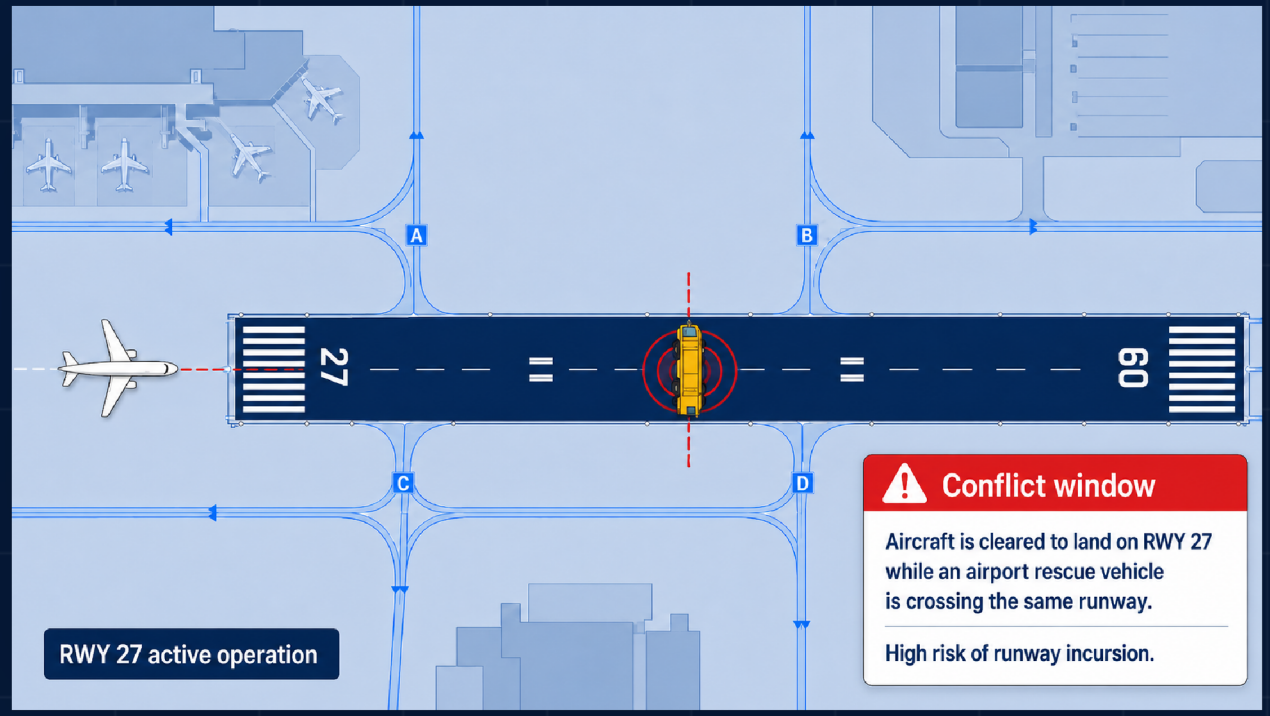
Explainable runway safety and air traffic control decision-support dashboard.

Helping air traffic teams see runway risk before it becomes a runway incident.

Team: Agentic Vibing

Focus: Advanced Technology and Innovation

Positioning: ATC support and training layer



THE PROBLEM | RUNWAY DECISIONS HAPPEN FAST

A runway conflict can form faster than a controller can rebuild the full picture.

The problem is not simply “many aircraft.” ATC workload rises when traffic, voice communication, weather, runway crossing, vehicle movement, and coordination happen at the same time.

What the controller is tracking

Aircraft approaching, landing, departing, or taxiing; active runway status; airport vehicles; runway crossing requests; weather and visibility; radio communication; readbacks; flight strips; and coordination across ATC units.

Where workload turns into risk

Critical information may be spread across different displays, voice communication, human memory, and procedure assumptions. When the time-to-conflict is short, even a small delay in recognizing the full picture can matter.

The gap we focus on

A landing or departing aircraft can conflict with a vehicle if clearance status, vehicle movement, and aircraft timing are not interpreted together. ATC FlowLens focuses on that exact intersection.

This is not about replacing controllers. It is about helping them see the runway picture sooner when workload is already high.

EVIDENCE | THIS RISK ALREADY SHOWS UP

The runway risk is already visible in safety data and recent incidents.

345
runway incursions across FAA Core 30 airports in FY2024, down from 362 in FY2023.

391
average annual incursions across Core 30 airports before the pandemic, FY2017-FY2019.

12
Core 30 airports exceeded their pre-pandemic incursion levels in FY2024.

DFW/BOS/ORD
had the highest Core 30 incursion counts in FY2024.

CASE	WHAT HAPPENED	WHAT IT PROVES FOR FLOWLENS
LATAM Peru Flight 2213, 2022	An A320neo collided with a fire truck on the runway during takeoff roll in Lima.	Aircraft-vehicle runway conflict can form during emergency vehicle activity and coordination gaps.
Austin-Bergstrom, 2023	A departure aircraft was cleared while another aircraft was on final approach in dense fog.	Low visibility and runway status awareness can reduce the safety margin quickly.
LaGuardia, 2026 preliminary	NTSB opened an investigation after a CRJ-900 collided with an ARFF vehicle on runway 4.	Vehicle tracking, last-second clearance, transponder coverage, and workload are not abstract concerns.

Sources: **FAA Air Traffic by the Numbers FY2024**, **NTSB DCA26MA161**, **BEA LATAM 2213 event page**, **AP/NTSB Austin coverage**.

THE GAP | TOOLS EXIST, CONTEXT STILL SPLITS

The tools exist, but the runway picture can still fragment under pressure.

What ATC already has

Modern ATC/ATM uses radar, ADS-B, MLAT, surface surveillance, electronic flight strips, A-SMGCS, RIMCAS/RMCA, STCA/MTCD, CPDLC/Data Comm, AMAN/DMAN, TFDM, and SWIM. These systems are important, but they are not equally available everywhere.

Why the gap still exists

Advanced surface tools such as ASDE-X, ASSC, A-SMGCS, and RIMCAS are valuable but can be expensive and complex. Voice communication also remains common, especially in tower and ground operations, so clearance state and runway timing can still require manual interpretation.

Where FlowLens fits

Signals already there

Aircraft, vehicles, runway, clearance, weather, communication logs.

→

FlowLens layer

Time-to-conflict, workload scoring, event explanation, replay.

→

Human stays in command

Controller remains the decision-maker. The tool supports awareness and training.

Sources: **ICAO Global Runway Safety Action Plan**, **MITRE surface safety prototypes**, **SKYbrary RIMCAS**, **FAA AIM ASDE-X/ASSC**.

THE SOLUTION | ATC FLOWLENS

ATC FlowLens turns runway movement, clearances, and workload into one explainable view.

The product does not automate ATC. It connects operational signals into a single timeline so teams can detect, explain, and replay runway risk.

Runway status at a glance

Tracks whether a runway is clear, occupied, being crossed, used for takeoff, or receiving an inbound landing aircraft.

Clearance memory

Maintains operational states such as cleared to land, cleared for takeoff, hold short, line up and wait, cleared to cross, and go around.

Time-to-conflict

Compares estimated aircraft touchdown or runway entry time with estimated vehicle runway clear time.

Communication in sequence

Organizes tower, pilot, and vehicle messages into an event history that can later support speech-to-text or readback mismatch detection.

Workload signal

Scores workload using traffic count, radio calls, active clearances, vehicle movements, weather, runway changes, and active alerts.

Replay that explains

Turns a risk event into a training object: what happened, when it happened, why the alert appeared, and what action category should be considered.

HOW IT WORKS | FROM TIMING TO WARNING

From aircraft and vehicle timing, FlowLens calculates the conflict before it becomes obvious.

1

Aircraft inbound

BAW123 is 1.8 NM final, speed 145 kt, cleared to land RWY 27.

2

Vehicle crossing

ARFF-2 is cleared or starts to cross RWY 27 during an active runway operation.

3

Timing check

The engine estimates touchdown in 42 seconds and vehicle clear time in 58 seconds.

4

Conflict window

Vehicle clear time is later than aircraft touchdown time. Predicted overlap: 16 seconds.

5

Clear explanation

The dashboard shows alert severity, reason, relevant objects, workload state, and replay timeline.

Example alert

HIGH RUNWAY CONFLICT RISK. BAW123 is cleared to land RWY 27. ARFF-2 is cleared to cross RWY 27. Vehicle clear time is later than aircraft touchdown time.

Processing pipeline

Inputs: aircraft, vehicle, runway, clearance, communication, weather. **Logic:** runway occupancy model, clearance state engine, time-to-conflict, workload score. **Output:** alert, reason, action category, replay timeline.

Conflict rule

```
IF aircraft.status = "cleared_to_land" AND vehicle.status = "cleared_to_cross" AND aircraft.runway = vehicle.runway AND vehicle_clear_time > aircraft_touchdown_time
THEN trigger HIGH runway conflict alert.
```

MVP | BUILD THE SAFETY LOGIC FIRST

The first prototype proves the runway safety logic without needing live ATC data.

What we can build now

One simulated airport layout with runway and taxiway, three to five aircraft, one to two airport vehicles, clearance and communication logs, weather or visibility status, rule-based conflict detection, time-to-conflict alerting, workload scoring, and replay mode.

Why it is buildable

It avoids live ATC data, regulatory integration, and certification dependency. The team can still demonstrate a credible safety concept using realistic scenarios and transparent logic.

Why it still creates value

Training centers and airport safety teams need explainable scenarios, not only raw alarms. FlowLens can become a reusable safety learning environment.

LAYER	WHAT WE BUILD NOW	WHAT IT CAN BECOME
Data	Scenario JSON and simulated positions.	ADS-B, airport operation data, weather feed, training simulator logs.
Logic	Rule-based time-to-conflict and workload score.	Richer prediction, scenario generator, human factors model.
Interface	Runway map, alert reason, event replay.	Training lab, safety analytics, research collaboration with ANSP/aviation school.

USERS & VALUE | START WHERE ADOPTION IS REALISTIC

The practical entry point is training, replay, and runway safety readiness.

Who feels this problem

Airport operators, Air Navigation Service Providers, ATC training centers, aviation academies, airport safety teams, and airport rescue and firefighting units.

Adoption path

1. Training and simulation tool. 2. Safety briefing and scenario replay product. 3. Airport operation support prototype. 4. Research collaboration. 5. Future certified decision-support exploration.

Training

Better runway conflict scenarios and controller workload awareness.

Review

Faster explanation of complex incident timelines and near-miss patterns.

Readiness

Safer coordination between tower, ground, emergency units, and airport operations.

Expansion

Speech-to-text, readback mismatch, richer workload analytics, and simulator integration.

A simulation-first product is a more credible starting point than claiming we can plug directly into certified ATC infrastructure.

IMPACT | WHY THIS FITS BUILD INDONESIA

ATC FlowLens fits BUILD by turning advanced technology into practical runway safety support.

IMPACT AREA	EXPECTED VALUE	BUILD INDONESIA ALIGNMENT
Runway safety	Helps users recognize runway conflict risk earlier by showing aircraft, vehicle, clearance, and timing in one view.	Aerospace & Defense: supports airport operational readiness and critical aviation safety.
Training and review	Makes training scenarios easier to explain and replay, especially for aircraft-vehicle conflict and communication sequence.	Advanced Technology and Innovation: uses digital decision support, simulated tracking, and explainable alerting.
Workload awareness	Gives a structured workload indicator based on traffic, radio calls, weather, vehicle movement, clearance load, and alerts.	Social Impact: safer operations protect passengers, crews, ground staff, and emergency units.
Operational flow	Future versions may reduce avoidable go-arounds, taxi delay, holding, and unnecessary fuel burn.	Sustainability: better runway flow can reduce waste in future operational use.

Main focus area

Advanced Technology and Innovation through decision support, prediction, workload scoring, and explainable runway safety visualization.

Additional value

Social Impact and Sustainability through safer runway operations, better coordination, and future reduction of avoidable fuel waste.

Why it can stand out

The idea is specific, evidence-backed, feasible as an MVP, and positioned carefully for a safety-critical aviation domain.

ROADMAP | START NARROW, GROW CAREFULLY

Start with a runway safety simulator. Grow toward an explainable ATC training platform.

Phase 1: Prove the scenario

Simulated runway map, aircraft and vehicle movement, clearance log, time-to-conflict alert, workload indicator, and replay timeline.

Phase 2: Add communication intelligence

ATC-style speech-to-text, readback mismatch detection, scenario generator, multi-runway support, and richer workload model.

Phase 3: Expand into training analytics

Integration with airport operation datasets, tower-ground training simulator, safety analytics dashboard, and research pathway toward certified support.

BUILD Indonesia can help us test the idea with the right mentors, aviation context, and product discipline.

Closing thought

ATC FlowLens is not trying to automate the controller. It gives controllers, trainers, and safety teams a clearer lens on runway flow, workload, and conflict risk when every second matters.